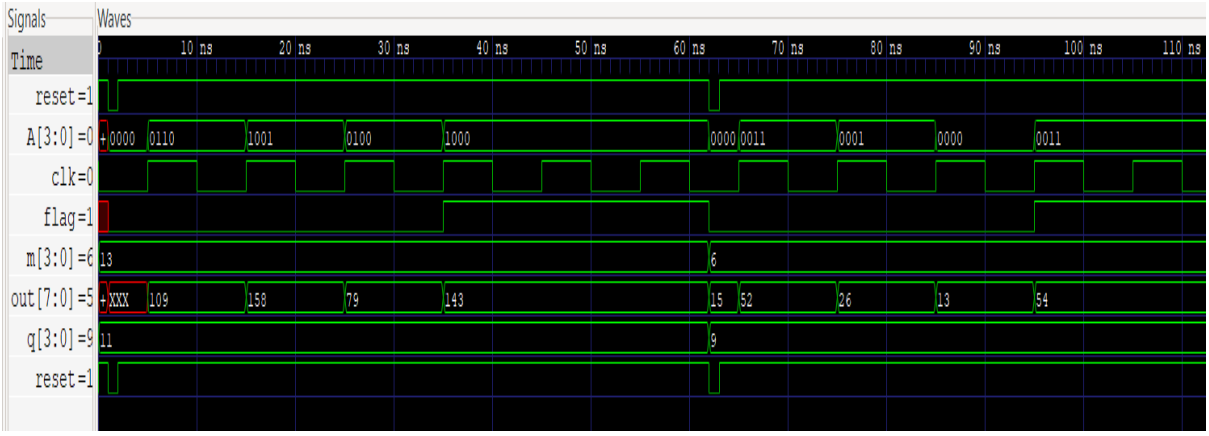


Multiplier

Works the same way as the schematic given in the question.

Posedge just used as a trigger to start addition then shifting then is reflected after the state change.(that is after adding and shifting)

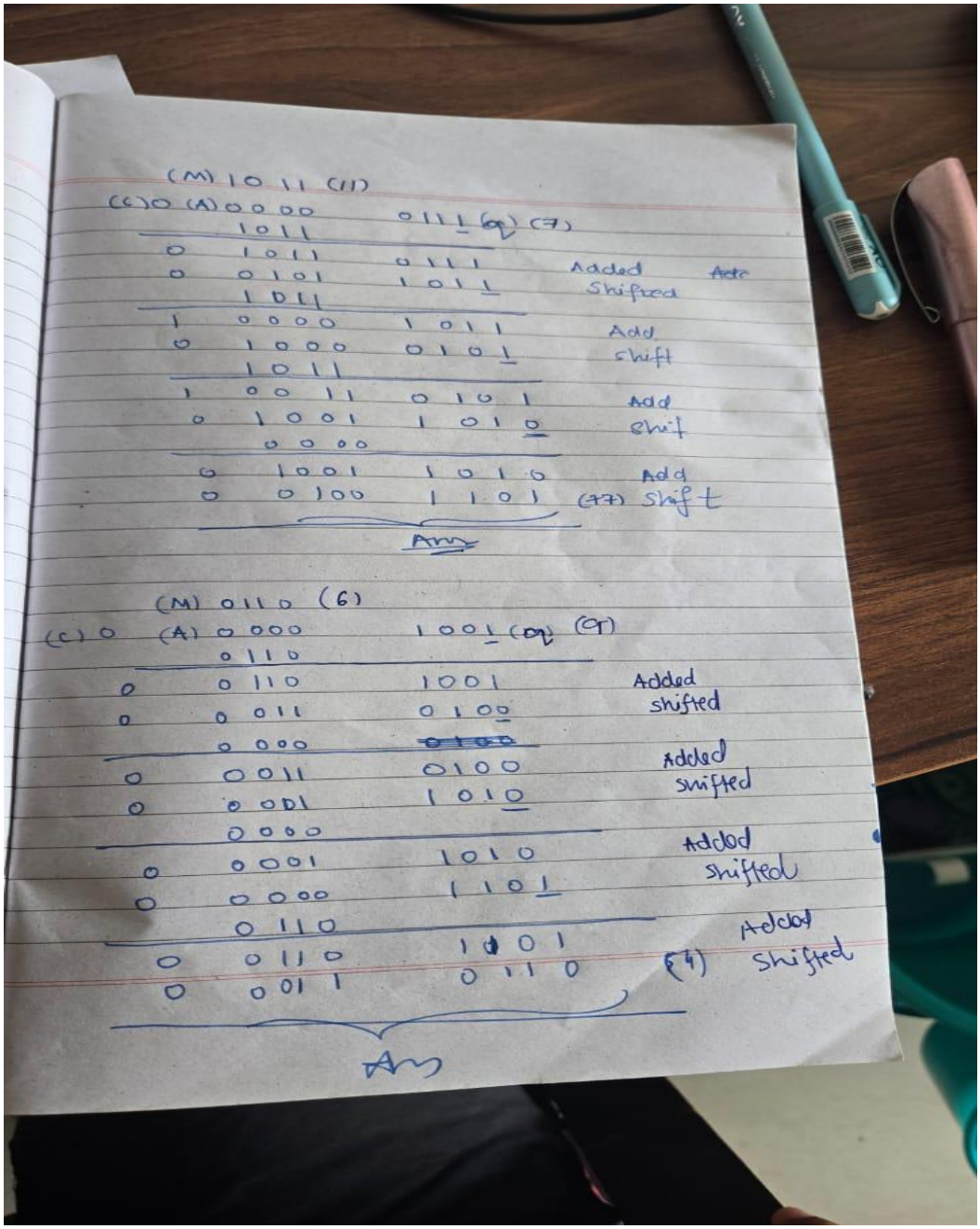
State just to keep track of the number of times the 2 processes take place.

All the major processes is commented in the code itself.

Reset acts as a phase where new values can be loaded.

Extra flag reg created to indicate answer is ready.

13x11 and 6x9 in the waveforms above.



(M) 1011 (11)

(C)	(A)			
0	0000	0111 (Q)	(7)	
	1011			
0	1011	0111		Added
0	0101	1011		Shifted
	1011			
1	0000	1011		Add
0	1000	0101		shift
	1011			
1	0011	0101		Add
0	1001	1010		shift
	0000			
0	1001	1010		Add
0	0100	1101	(77)	shift

Ans

(M) 0110 (6)

(C)	(A)			
0	0000	1001 (Q)	(9)	
	0110			
0	0110	1001		Added
0	0011	0100		shifted
	0000	~~0100~~		
0	0011	0100		Added
0	0001	1010		shifted
	0000			
0	0001	1010		Added
0	0000	1101		shifted
	0110			
0	0110	1001		Added
0	0011	0110	(54)	shifted

Ans